



**The research of homogeneity scale of the
universe.**

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Advisor: Jailson Souza de Alcaniz

29 de agosto de 2024

Academic Status

Research

Results

Reference

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Academic Status

– Courses

Ano/Semestre	Disciplina	Conceito	Créditos
2023/1	Stellar Evolution	A	4
2023/1	Data Analysis Tools in Cosmology	A	2
2023/2	Radiative processes in astrophysics	A	4
2023/2	Astrostatistics	A	4

– Exam of Qualify

Approved on June 13, 2024. Theme: Linear perturbation theory in cosmology.

– Events and other activities

- ▶ Weekly meetings: National Observatory Cosmology Seminars & Journal Club.
- ▶ Participation in ICTP Winter School on Cosmology in Argentina in 2022.
- ▶ Presentation at the Annual Postgraduate Seminars in Astronomy at the National Observatory (ON) on July, 2023.
- ▶ Presentation at the III SBF Spring Meeting on September, 2023.

– Publication

Shao X, Gonçalves R S, Bengaly C A P, et al. Can the angular scale of cosmic homogeneity be used as a cosmological test ?[J]. The European Physical Journal C, 2024, 84(7): 655.

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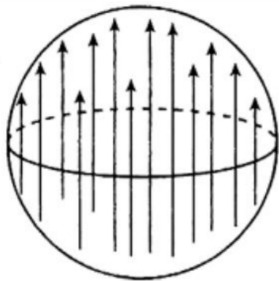
Research

Research:

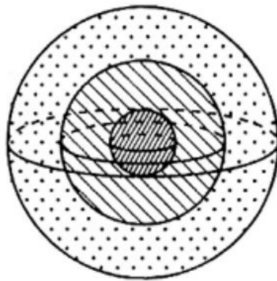
Investigate theoretical and observational aspects of cosmic homogeneity and study the cosmic homogeneity as a cosmological probe (Shao X et al, EPJC, 2024).

- ▶ Standard cosmology: General Relativity + Cosmology Principle
- ▶ General Relativity: describe gravity as a geometric property of space and time
- ▶ Cosmology Principle: homogeneity and isotropy
- ▶ Homogeneity: No special places in the Universe
- ▶ Isotropy: No special directions in the Universe

Introduction



Homogeneous
Not isotropic

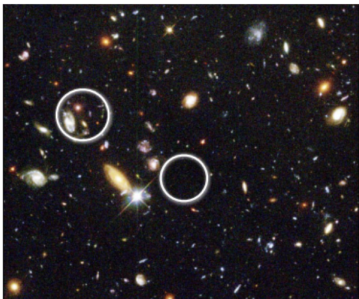


Isotropic
Not homogeneous

- ▶ no special places, but special direction;
- ▶ no special direction, but special places.

Introduction

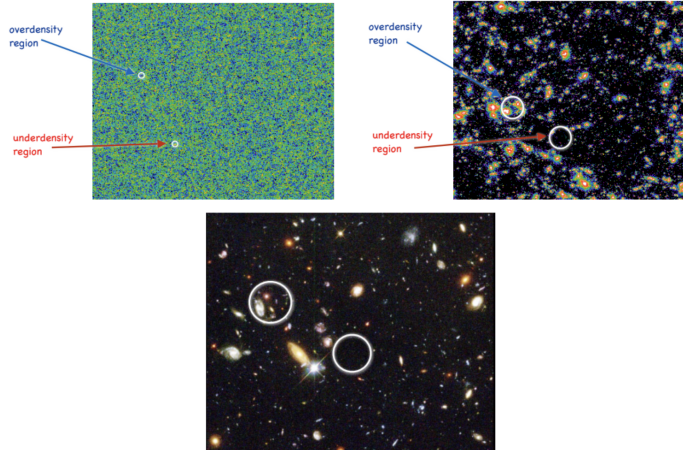
- ▶ Perfectly homogeneous and isotropic $\xrightarrow{\text{True ?}}$ no formation of structures.
- ▶ Our universe contains interesting objects $\rightarrow$ planets and stars, galaxies and clusters of galaxies.



- ▶ Not perfectly homogeneous and isotropic.

Introduction

- Evolution of inhomogeneities due to gravitational instability



Introduction

- ▶ Whether there is still a scale that our universe is homogeneous ?
- ▶ If the scale exists, how big it should be ?

Introduction

- ▶ Many research and textbook refer to: The Universe is homogeneous on scales larger than 100 Mpc/h (Megaparsec per h).
- ▶ The universe is expanding. The homogeneity scale evolves with redshift. Smaller at early times; Bigger at later times;

Can the homogeneity scale be used as a cosmological test?

Method

- The scaled count-in-spheres:

The the number ratio of the neighboring galaxies in a 3D-sphere of radius r around a given data point and the one in a random catalogue:

$$\mathcal{N}(< r) = \frac{N(< r)}{N_{\text{random}}(< r)} \quad (1)$$

$$\begin{array}{ll} N(< r) \propto r^{\mathcal{D}_2} & \text{Real galaxies} \\ N_{\text{random}}(< r) \propto r^3 & \text{Random galaxies} \end{array} \quad (2)$$

- Fractal dimension:

$$\mathcal{D}_2(r) = 3 + \frac{d \ln \mathcal{N}(< r)}{d \ln r} \quad (3)$$

Method

- ▶ We define the homogeneity scale R_H as the scale at which the Universe deviates at 1% from homogeneity.
- ▶ Fractal dimension:

$$\mathcal{D}_2(r_H) \rightarrow 2.97 \quad (4)$$

Can the spatial homogeneity scale be used as a cosmological test?

Method

Can the homogeneity scale be used as a standard ruler?

Savvas Nesseris^{*} and Manuel Trashorras[†]

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Cantoblanco, 28049 Madrid, Spain*

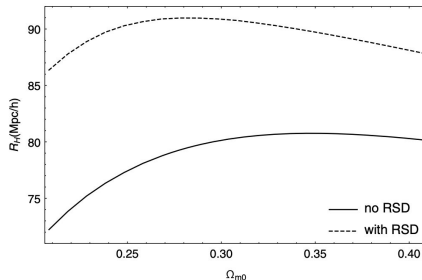


(Received 21 January 2019; published 29 March 2019)

The Universe on comoving scales larger than 100 Mpc/h is assumed to be statistically homogeneous, with the transition scale where we have 1% deviation from homogeneity being known as the homogeneity scale R_H . The latter was recently proposed in Ntelis *et al.* [[J. Cosmol. Astropart. Phys. 12 \(2018\) 014](#)] to be a standard ruler. In this paper we perform a comparison between the baryon acoustic oscillations and the homogeneity scales R_H , assuming a spatially universe, linear cosmological perturbation theory and the cosmological constant cold dark matter (Λ CDM) model. By direct theoretical calculations, we demonstrate that the answer to the question of whether the homogeneity scale can be used as a standard ruler is clearly negative due to the nonmonotonicity of the homogeneity scale R_H with the matter density parameter Ω_{m0} of the Λ CDM model. Furthermore, we also consider the effect of redshift-space distortions and we find that they do not break the degeneracies but instead only change the value of R_H by a few percent. Finally, we also confirm our findings with the help of an N -body simulation.

Can the spatial homogeneity scale be used as a cosmological test?

- Fractal dimension: $\mathcal{D}_2(r_H) \rightarrow 2.97$



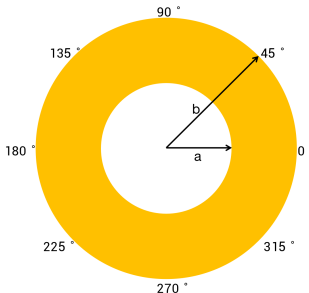
The spatial homogeneity scale is not monotonic with matter density and can't be used as a standard ruler.

Method

- ▶ How about angular homogeneity scale ?
- ▶ Can angular homogeneity scale be used as a cosmological test?
- ▶ Model-independent quantity

Fractal dimension

- The scaled count-in-spheres:



- The number ratio of neighboring galaxies in a 2D-sphere of radius θ around a given data point and the one in a random catalogue:

$$\mathcal{N}(<\theta) = \frac{N(<\theta)}{N_{\text{random}}(<\theta)}$$

- The scaled count-in-spheres:

$$\mathcal{N}(< \theta) = \frac{N(< \theta)}{N_{\text{random}}(< \theta)} \quad (6)$$

$$N(< \theta) \propto \theta^{\mathcal{D}_2} \text{ Real galaxies} \quad (7)$$

$$N_{\text{random}}(< \theta) \propto \theta^2 \text{ Random galaxies} \quad (8)$$

- Fractal dimension:

$$\mathcal{D}_2(\theta) = 2 + \frac{d \ln \mathcal{N}(< \theta)}{d \ln \theta} \quad (9)$$

Fractal dimension

- ▶ We define the angular homogeneity scale θ_H as the scale at which the Universe deviates at 1% from homogeneity
- ▶ Fractal dimension in 2D:

$$D_2(\theta_H) \rightarrow 1.98 \quad (10)$$

Fractal dimension

Angular homogeneity scale is model-independent :

- Observational data $\rightarrow$ Landy-Szalay(LS) estimator

$$\hat{\omega}_{ls}(\theta) = \frac{DD(\theta) - 2DR(\theta) + RR(\theta)}{RR(\theta)}, \quad (11)$$

where $DD(\theta)$, $DR(\theta)$, and $RR(\theta)$ are the numbers of pairs of data as a function of separations θ in data-data, data- random and random-random catalogues.

- Fractal dimension:

$$\mathcal{D}_2(\theta) = 2 + \frac{d \ln \mathcal{N}(< \theta)}{d \ln \theta} \rightarrow 1.98 \quad (12)$$

Fractal dimension

- Theoretical calculation:

Matter power spectrum(CAMB) $\rightarrow$ two point correlation function

- Fractal dimension:

$$\mathcal{D}_2(\theta) = 2 + \frac{d \ln \mathcal{N}(< \theta)}{d \ln \theta} \rightarrow 1.98 \quad (13)$$

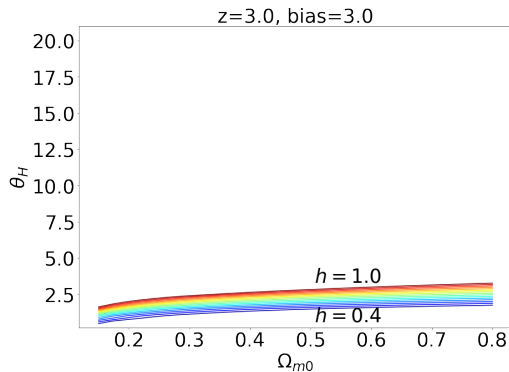
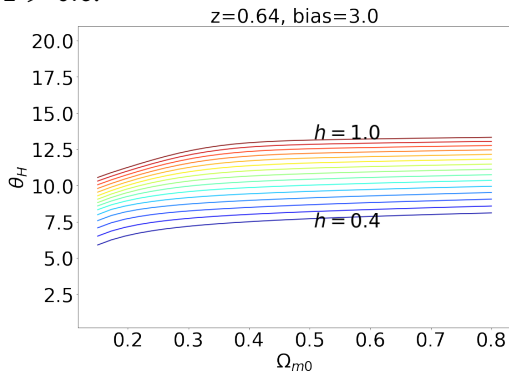
Can angular homogeneity scale be used as a cosmological test?

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Results

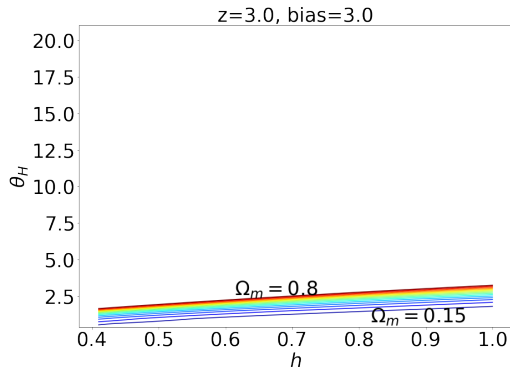
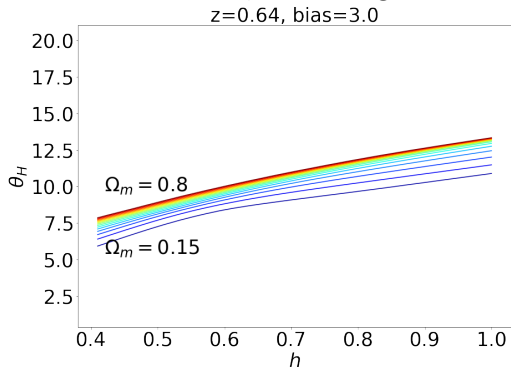
Results

- That angular homogeneity scale θ_H is a monotonically increasing function for Ω_{m0} for $z > 0.6$.



Results

- θ_H is a monotonically increasing function for H_0 for $z > 0.6$.



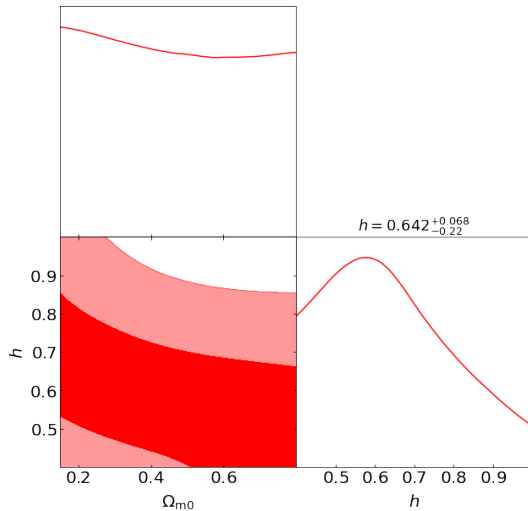
- The angular homogeneity scale can be used as a cosmological test. And it very sensitive to the change of redshift.

Results

Use the three currently available measurements of θ_H from Dr16 eBOSS obtained by "Andrade U, Gonçalves R S, Carvalho G C, et al. The angular scale of homogeneity with SDSS-IV DR16 luminous red galaxies[J]"

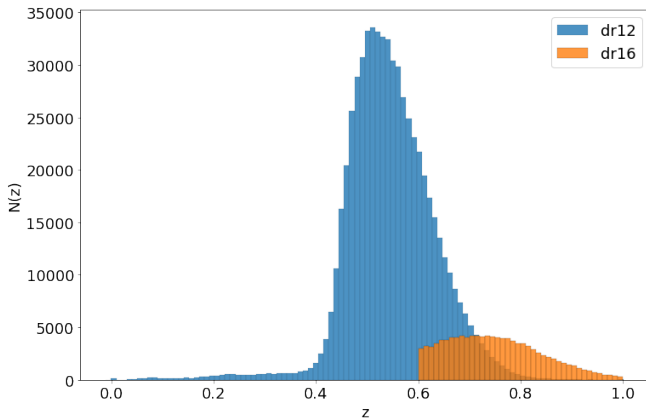
z	$\bar{z}$	N_{gal}	θ_H (deg)
0.67 - 0.68	0.675	4209	7.57 ± 2.91
0.70 - 0.71	0.705	4073	7.49 ± 2.63
0.73 - 0.74	0.735	4210	8.88 ± 2.81

Results



Results

Data: LRGs from BOSS DR12 and eBOSS DR16 catalogues (redshift range: $0.46 < z < 0.74$).



Results

The redshift bins adopted in our analysis from BOSS (DR12):

z	$\bar{z}$	N_{gal}
0.46-0.47	0.465	22551
0.47-0.48	0.475	27319
0.48-0.49	0.485	29251
0.49-0.50	0.495	31763
0.50-0.51	0.505	33107
0.51-0.52	0.515	32887
0.52-0.53	0.525	32794
0.53-0.54	0.535	31995
0.54-0.55	0.545	31355
0.55-0.56	0.555	29486
0.56-0.57	0.565	28995
0.57-0.58	0.575	25289
0.58-0.59	0.585	23997
0.59-0.60	0.595	22568
0.60-0.61	0.605	20594
0.61-0.62	0.615	18799

Results

The redshift bins adopted in our analysis from eBOSS (DR16), along with the redshift bin means and corresponding number of LRGs in each bin:

z	$\bar{z}$	N_{gal}
0.67-0.68	0.675	4073
0.68-0.69	0.685	4222
0.69-0.70	0.695	4064
0.70-0.71	0.705	4209
0.71-0.72	0.715	4239
0.72-0.73	0.725	4084
0.73-0.74	0.735	4210

Results

The redshift and the respective observational angular homogeneity scale obtained from the DR12 data set:

$\bar{z}$	$\theta_H(\text{degree})$
0.465	10.4 ± 2.47
0.475	10.43 ± 2.58
0.485	9.70 ± 2.73
0.495	9.08 ± 2.50
0.505	9.14 ± 2.48
0.515	9.28 ± 2.76
0.525	9.57 ± 2.39
0.535	9.80 ± 2.67
0.545	9.72 ± 2.69
0.555	9.17 ± 2.54
0.565	9.35 ± 2.77
0.575	9.07 ± 2.75
0.585	8.31 ± 2.63
0.595	8.88 ± 2.87
0.605	8.54 ± 2.90
0.615	8.36 ± 2.90

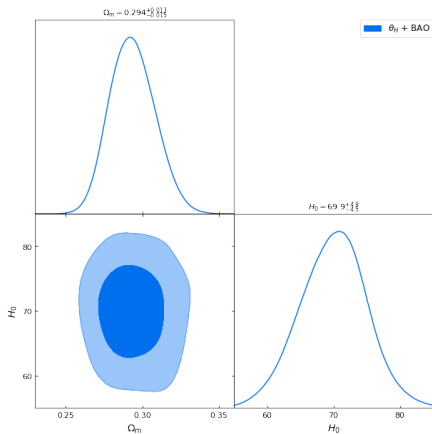
Results

The redshift and the respective observational angular homogeneity scale obtained from the DR16 data set:

$\bar{z}$	$\theta_H^{boot}(\text{degree})$
0.675	7.49 ± 2.83
0.685	8.53 ± 2.58
0.695	8.07 ± 2.59
0.705	7.36 ± 2.49
0.715	6.92 ± 2.28
0.725	8.12 ± 2.04
0.735	8.89 ± 2.90

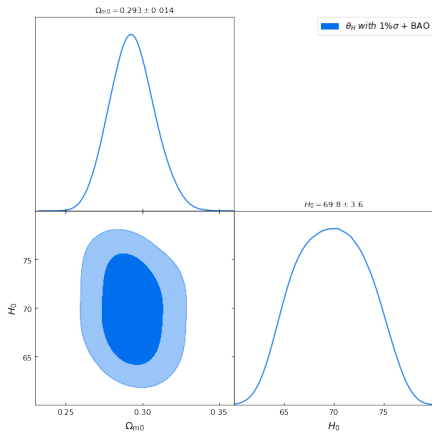
Results

The results of angular homogeneity scale combining with BAO by fixing bias:



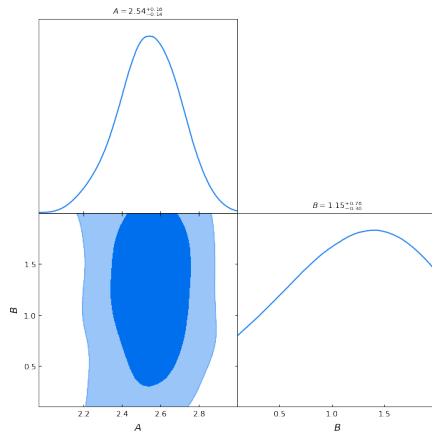
Results

Simulate θ_H data with $1\%\sigma$. Marginalized posteriors on Ω_{m0} and H_0 combining with BAO.



Results

The results of bias $\{A, B\}$ obtained by fixing Ω_{m0} and H_0 at the value of PLANCK 2018:



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Reference

- [1] Rodrigo S. Gonçalves, Gabriela C. Carvalho, Uendert Andrade, Carlos A. P. Bengaly, Joel C. Carvalho, and Jailson Alcaniz. Measuring the cosmic homogeneity scale with SDSS-IV DR16 Quasars. JCAP, 03:029, 2021
- [2] R. S. Gonçalves, G. C. Carvalho, C. A. P. Bengaly, J. C. Carvalho, A. Bernui, J. S. Alcaniz, and R. Maartens. Cosmic homogeneity: a spectroscopic and model-independent measurement. Mon. Not. Roy. Astron. Soc., 475(1):L20–L24, 2018.
- [3] Uendert Andrade, Rodrigo S. Gonçalves, Gabriela C. Carvalho, Carlos A. P. Bengaly, Joel C. Carvalho, and Jailson Alcaniz. The angular scale of homogeneity with SDSS-IV DR16 luminous red galaxies. JCAP, 10:088, 2022.

[4] Savvas Nesseris and Manuel Trashorras. Can the homogeneity scale be used as a standard ruler? Phys. Rev. D, 99(6):063539, 2019.

[5] Saadeh, D., Feeney, S. M., Pontzen, A., Peiris, H. V., McEwen, J. D. (2016). How isotropic is the Universe?. Physical review letters, 117(13), 131302.

Obrigada!